EXPERIMENT NO. 1

Estimation of Characteristic Impedance (Z_0) and Propagation Constant (γ) of a Transmission Line from Primary Constants

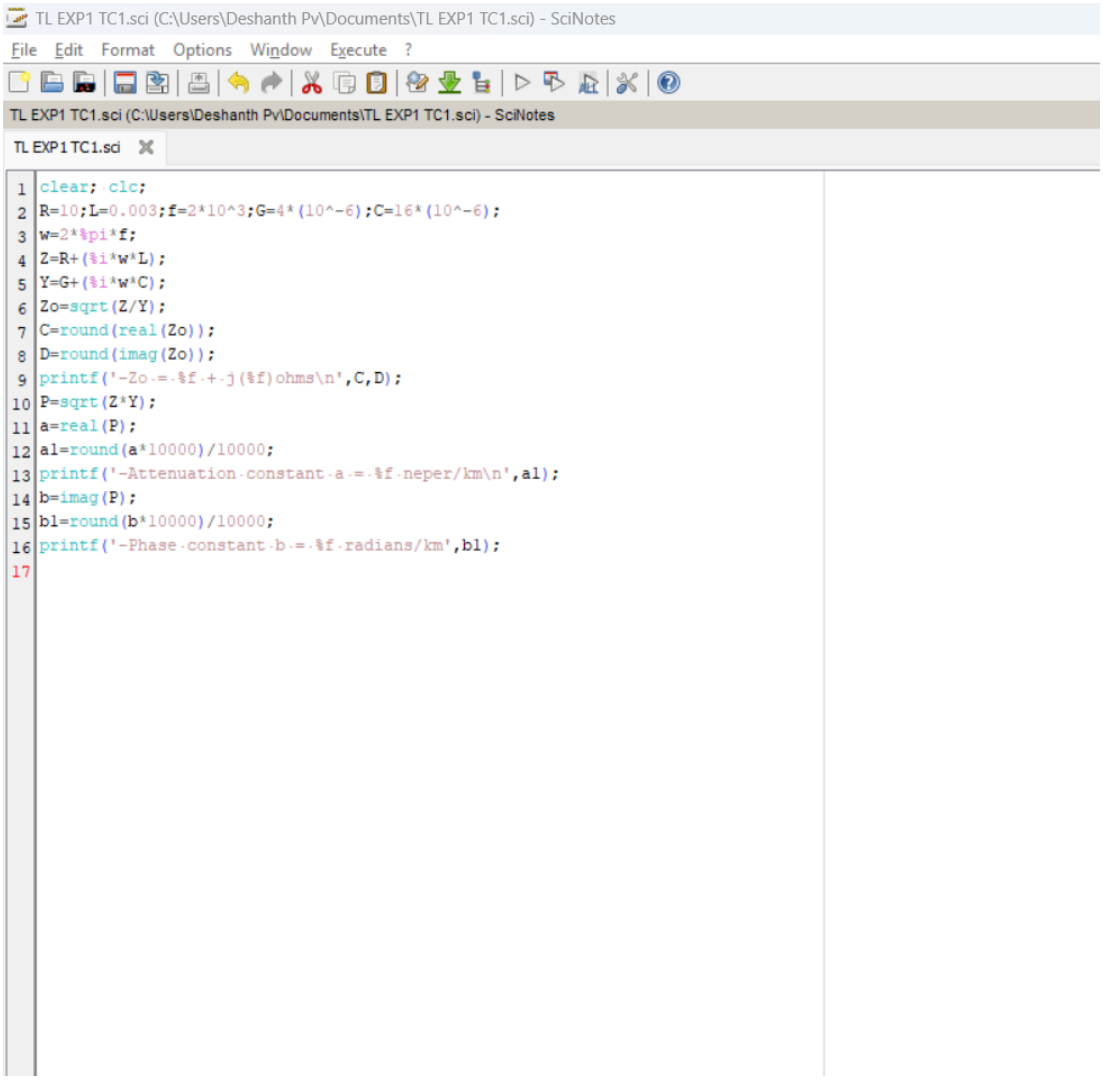
AIM

Investigating the significance of Characteristic Impedance (Z_0) and Propagation Constant (γ) of a Transmission Line with Resistance = 10 Ω/km, Capacitance = 8 μF/km, Inductance = 3.5 mH/km, Conductance = 4 μmho/km and Frequency = 2 kHz.

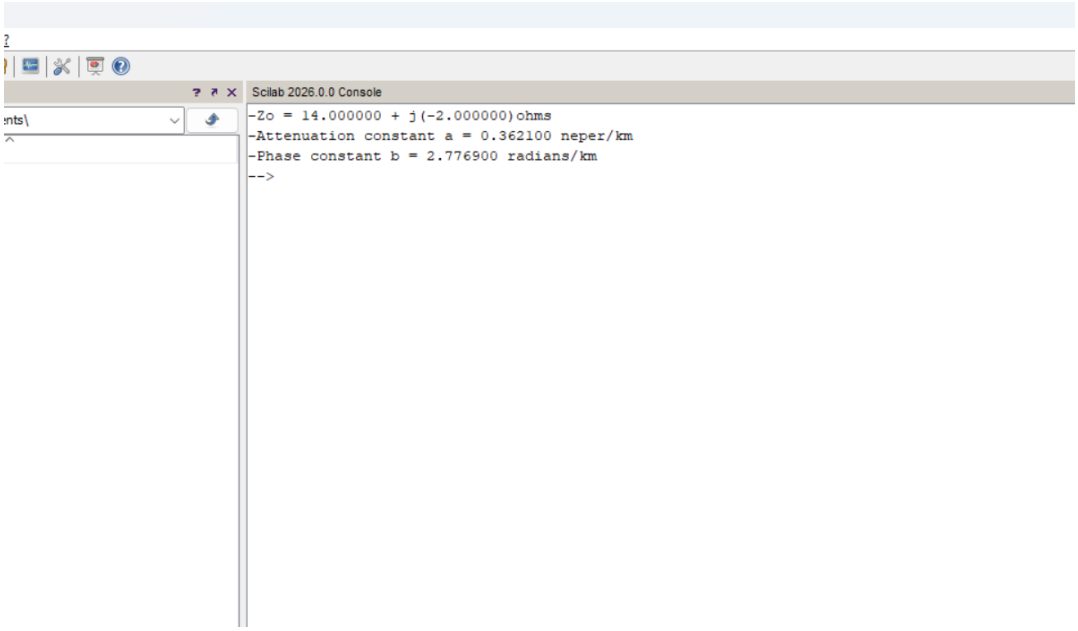
APPARATUS

SCILAB Version 6.1.1

TEST CASE 1:

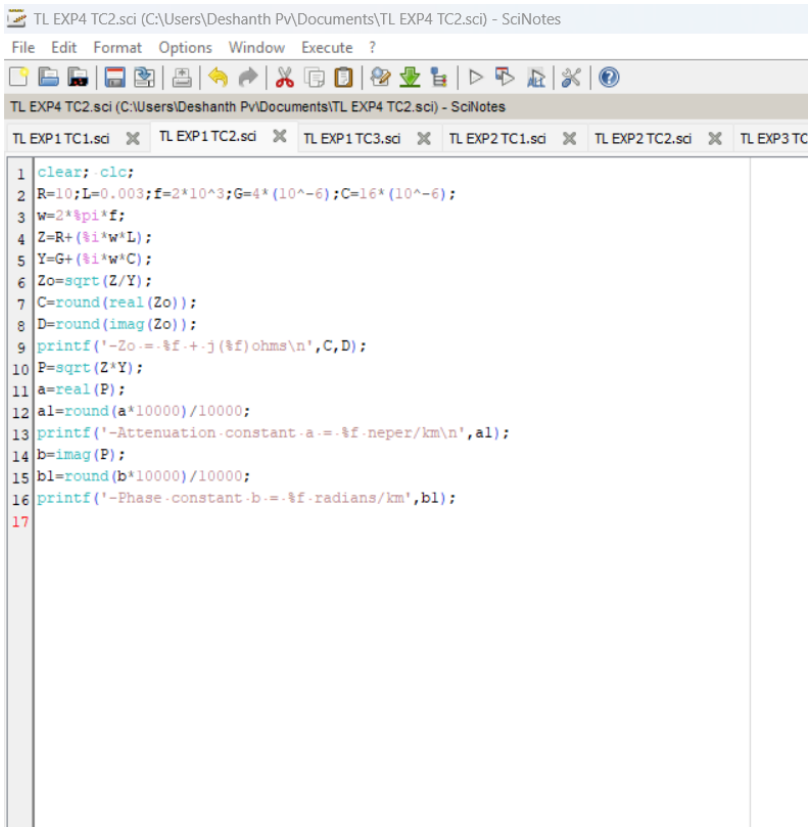


OUTPUT

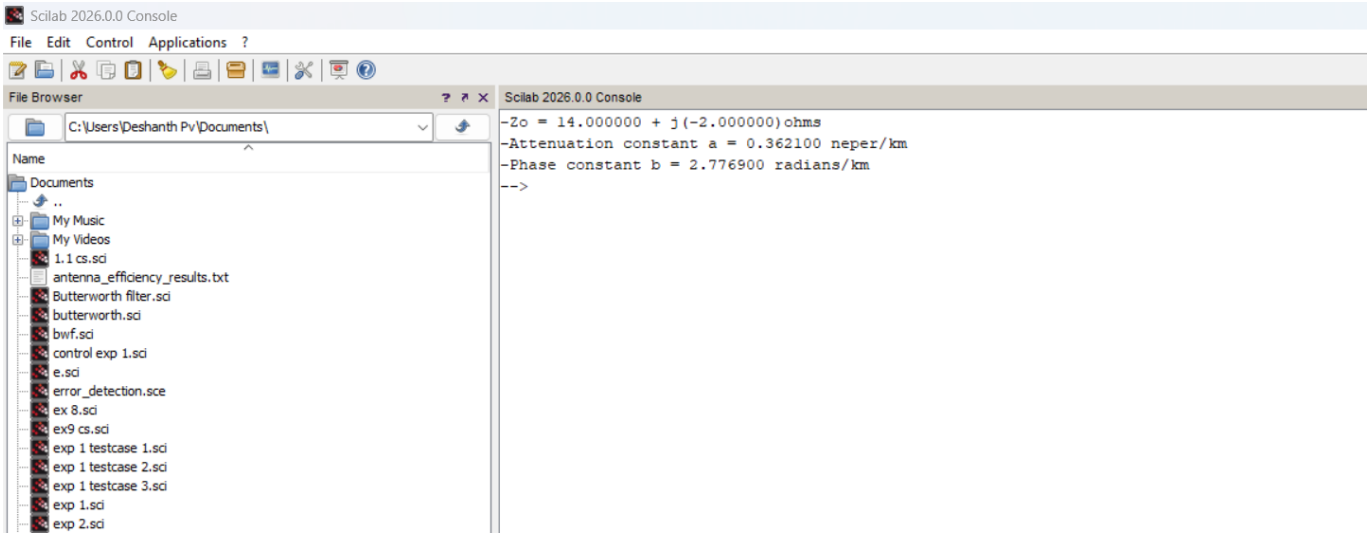


Quantity	Output
Characteristic impedance, Z_0	14.000000 + j(-2.000000) Ω
Attenuation constant, α	0.362100 neper/km
Phase constant, β	2.776900 radians/km

TEST CASE 2:

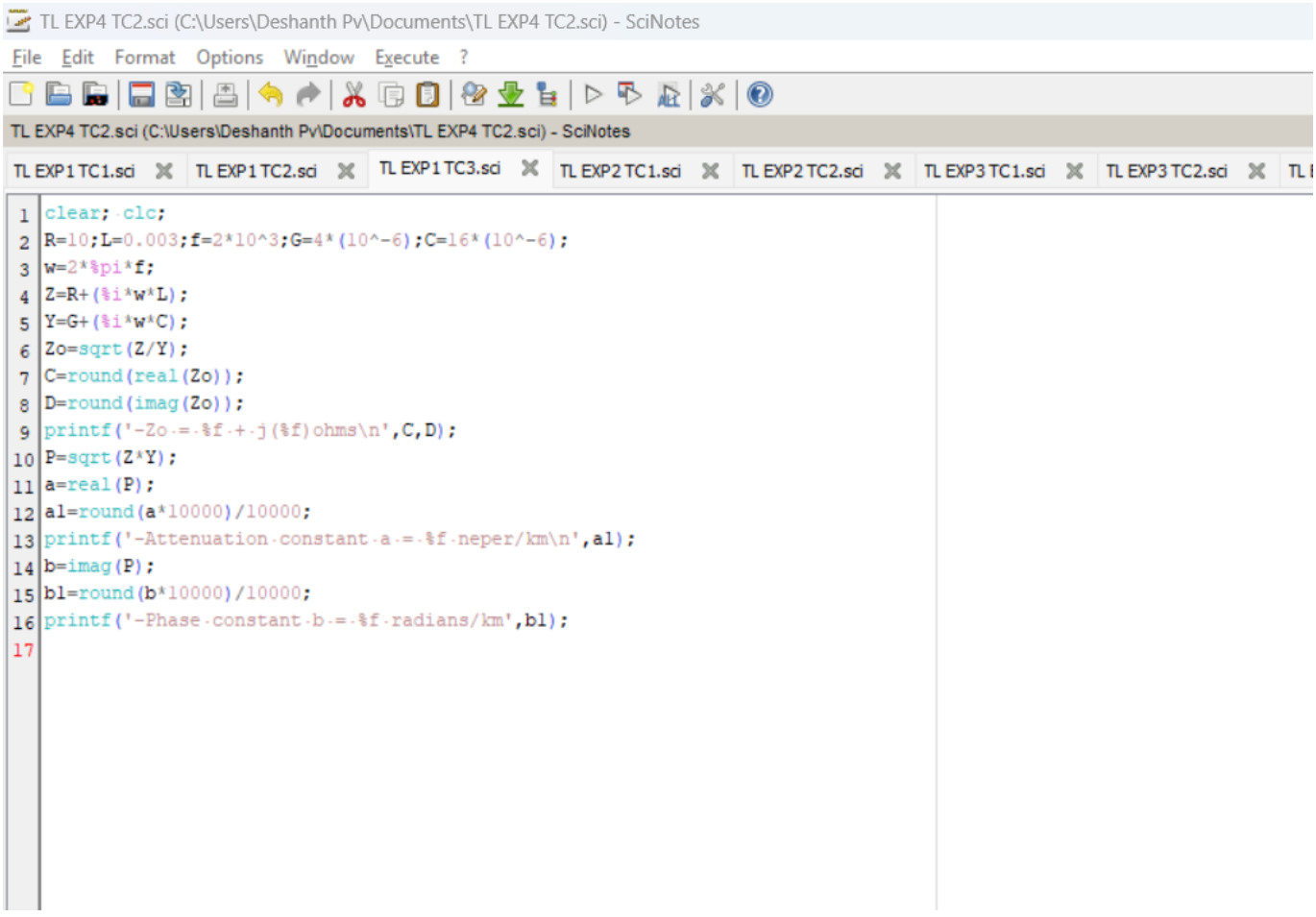


OUTPUT:

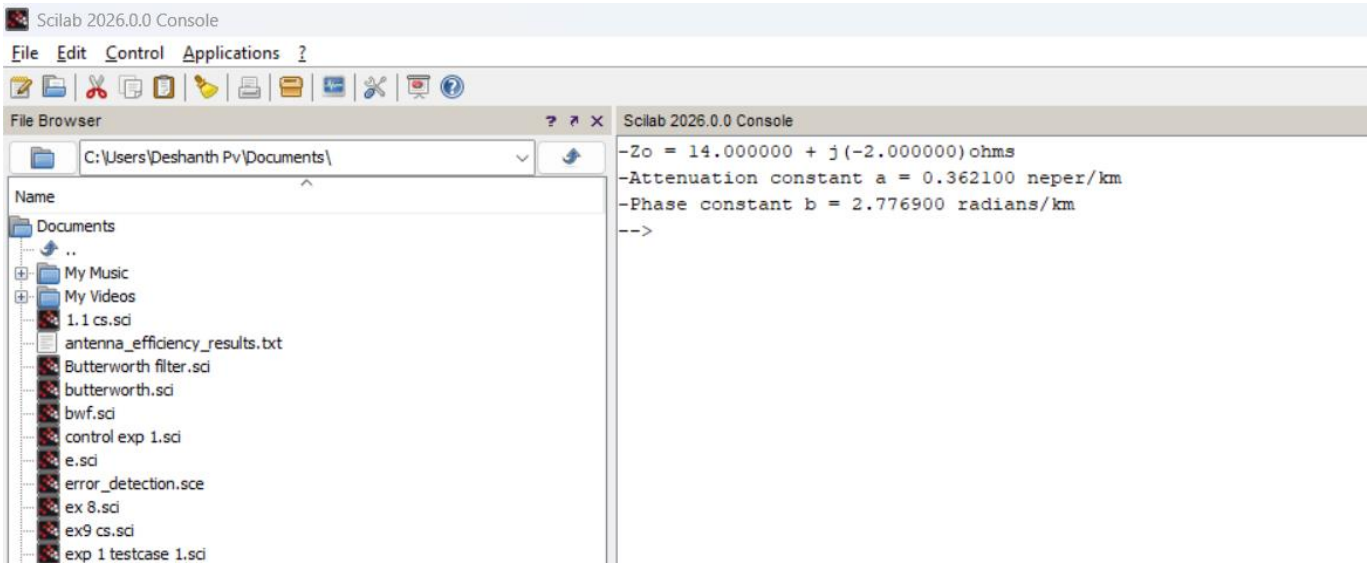


Quantity	Output
Characteristic impedance, Z_0	14.000000 + j(-2.000000) Ω
Attenuation constant, α	0.362100 neper/km
Phase constant, β	2.776900 radians/km

TEST CASE 3:



OUTPUT:



Simulation output obtained from the provided experiment record:

Quantity	**Output**
Characteristic impedance, Z_0	14.000000 + j(-2.000000) Ω
Attenuation constant, α	0.362100 neper/km
Phase constant, β	2.776900 radians/km

RESULT

The characteristic impedance and propagation constant of a transmission line are calculated using SCILAB, and the impact of primary constants on secondary constants is investigated.